THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Fractions in Disguise** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A machine worth ₹ 80000 depreciates at 20% per annum. Its value after 2 years (in ₹) is: 1

(a) 56320 **(b)** 51200 **(c)** 48000 **(d)** 46080

2. A price is first increased by 10% and then decreased by 25%. The net change (as percent, + for increase, − for decrease) is: 1

(a) −15% **(b)** −15.75% **(c)** −17.5% **(d)** −19.25%

3. The CI on a sum for 2 years at 10% p.a. is ₹ 840. The sum (in ₹) is: 1

(a) 4000 **(b)** 3600 **(c)** 4400 **(d)** 8000

4. A trader marks his goods 50% above cost and allows a discount of 20%. His gain percent is: 1

(a) 20% **(b)** 30% **(c)** 18% **(d)** 22%

5. If the difference between the CI and the SI on a sum for 2 years at 8% is ₹ 96, the sum (in ₹) is: 1

(a) 16500 **(b)** 13500 **(c)** 30000 **(d)** 15000

6. A sum triples in 8 years at simple interest. The rate of interest per annum is: 1

(a) 22% **(b)** 25% **(c)** 50% **(d)** 28%

7. Two articles are sold at ₹ 3000 each, one at a gain of 15% and the other at a loss of 15%. The overall result is a loss of (in %): 1

(a) 15% **(b)** 7.5% **(c)** 0% **(d)** 2.25%

8. The population of a town is 62500 and it grows at 20% per annum. After 2 years it will be: 1

(a) 99000 **(b)** 87500 **(c)** 81000 **(d)** 90000

9. **Assertion (A):** The compound interest on ₹1,000 at 10% per annum for 2 years is ₹210. 1
Reason (R): Compound interest and simple interest on the same sum for 2 years are always equal.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** An item bought for ₹500 and sold at a loss of 20% is sold for ₹600. 1
Reason (R): Loss percent is calculated on the cost price.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The cost of an item is ₹ 2400. It is marked 25% above the cost price and sold at 20% discount. Find the selling price. 2

12. Rahul invests ₹ 4000 at 10% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

13. A number is increased by 15% and the result is decreased by 10%. Is the final number more or less than the original? By what percent? 2

14. The cost of an item is ₹ 3600. It is marked 50% above the cost price and sold at 30% discount. Find the selling price. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A merchant sells two items at ₹ 1950 each, one at a gain of 25% and the other at a loss of 25%. Find the overall gain or loss in rupees. 3

16. The value of a car depreciates by 20% every year. If its present value is ₹ 250000, find its value after 3 years. 3

17. A shopkeeper buys an article at ₹ 2000, marks it up by 60% and gives a discount of 30%. Find his gain or loss percent. 3

OR

A merchant sells two items at ₹ 3360 each, one at a gain of 40% and the other at a loss of 40%. Find the overall gain or loss in rupees. 3

18. The value of a car depreciates by 10% every year. If its present value is ₹ 800000, find its value after 3 years. 3

OR

A shopkeeper buys an article at ₹ 500, marks it up by 30% and gives a discount of 10%. Find his gain or loss percent. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A shopkeeper in Surat buys 40 notebooks at ₹25 each. He sells 30 of them at ₹35 each and the remaining at ₹20 each. 5

(i) Find the total cost price.

(ii) Find the selling price of the 30 notebooks.

(iii) Find the selling price of the remaining notebooks.

(iv) Find his overall profit or loss and the percent.

20. A sum of ₹ 40000 is invested for 3 years at 10% p.a. compounded annually. Find the interest for each of the three years separately and verify that their total is the compound interest. 5

OR

A shopkeeper marks an article 80% above its cost price. During a sale he gives a discount of 30% and still earns a profit of ₹ 260. Find the cost price. 5

END OF QUESTION PAPER — Total: 40 Marks